

General Romberg FORTRAN95 code - RomberG

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Author's note: This documentation and software is based on the paper “M. Medved et al. / Journal of Molecular Structure: THEOCHEM 847 (2007) 39–46”^[1]

1 A brief summary on the Rutishauser-Romberg methodology and its generalization

The original Romberg methodology was originally developed for numerical integration to improve the accuracy of the trapezoidal rule by linearly combining lower order trapezoids.

Following Romberg's work, Rutishauser proposed using the latter for numerical differentiation by exploiting an error, as in the order of magnitude, in the central differences formula. For it let us consider the Taylor series expansion of a function $f(x)$ but evaluated at a nearby point displaced by h , *i.e.* $f(x+h)$ and $f(x-h)$:

$$\begin{cases} f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f^{(3)}(x) + \frac{h^4}{4!}f^{(4)}(x) + \dots \\ f(x-h) = f(x) - hf'(x) + \frac{h^2}{2!}f''(x) - \frac{h^3}{3!}f^{(3)}(x) + \frac{h^4}{4!}f^{(4)}(x) + \dots \end{cases} \quad (1)$$

And subtracting $f(x-h)$ from $f(x+h)$ yields:

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{2h^3}{3!}f^{(3)}(x) + \frac{2h^5}{5!}f^{(5)}(x) + \dots \quad (2)$$

$$D(h) = \frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{h^2}{3!}f^{(3)}(x) + \frac{h^4}{5!}f^{(5)}(x) + \dots \quad (3)$$

$$D(h) \simeq f'(x) + \mathcal{O}(h^2) \quad (4)$$

Consequently, having an error from the second decimal. On the other hand, if we consider $D(h)$ expansions now using a $\frac{h}{2}$ step we can redefine the definition of the first derivative with a lower error:

$$D(h) = f'(x) + Ah^2 + Bh^4 + \dots \quad (5)$$

$$D(h/2) = f'(x) + A\frac{h^2}{4} + B\frac{h^4}{16} + \dots \quad (6)$$

We can multiply by four the $D(h/2)$ expansion and subtract the $D(h)$ one so the first derivative of $f(x)$ is defined as:

$$4D(h/2) = 4f'(x) + Bh^4 + \dots \quad (7)$$

$$D(h) = f'(x) + Ah^2 + Bh^4 + \dots \quad (8)$$

$$4D(h/2) - D(h) = 3f'(x) + \mathcal{O}(h^4) \quad (9)$$

$$\frac{4D(h/2) - D(h)}{3} = f'(x) + \mathcal{O}(h^4) \quad (10)$$

and reducing the magnitude of the error to $\mathcal{O}(h^4)$. Then, combining $f(x)$ computed at different multiples of h the previous expression can be iterated to reduce higher order contamination, known as the Romberg triangle:

$$D^{p,k} = \frac{4^p D^{p-1,k} - D^{p-1,k+1}}{4^p - 1} \quad p = 1, 2, \dots \quad (11)$$

where p is the number of the Romberg iteration, k is related to the distance of the initial points from $x=0$, and D the derivative of our interest.

Using the previous methodology of subtracting Taylor series expansions it can be derived the following set of formulae for up to the fourth-order derivative:

$$f^{(1)} \approx \frac{f(2^k h) - f(-2^k h)}{2^{k+1} h} \quad (12)$$

$$f^{(2)} \approx \frac{f(2^k h) - 2f(0) + f(-2^k h)}{(2^k h)^2} \quad (13)$$

$$f^{(3)} \approx \frac{f(2^{k+1} h) - 2f(2^k h) + 2f(-2^k h) - f(-2^{k+1} h)}{2(2^k h)^3} \quad (14)$$

$$f^{(4)} \approx \frac{f(2^{k+1} h) - 4f(2^k h) + 6f(0) - 4f(-2^k h) - f(-2^{k+1} h)}{(2^k h)^4} \quad (15)$$

where these formulae can only be applied for step sizes increasing in powers of two. The general set of formulae applicable for step sizes $a \in [1, 2]$, and the ones implemented in this code and computed on the fly based on the data, are the following:

$$f^{(1)} \approx \frac{f(a^k h) - f(-a^k h)}{2a^k h} \quad (16)$$

$$f^{(2)} \approx \frac{f(a^k h) - 2f(0) + f(-a^k h)}{(a^k h)^2} \quad (17)$$

$$f^{(3)} \approx 3 \cdot \frac{f(a^{k+1} h) - af(a^k h) + af(-a^k h) - f(-a^{k+1} h)}{a(a^2 - 1)(a^k h)^3} \quad (18)$$

$$f^{(4)} \approx 12 \cdot \frac{f(a^{k+1} h) - a^2 f(a^k h) + 2(a^2 - 1)f(0) - a^2 f(-a^k h) - f(-a^{k+1} h)}{a^2(a^2 - 1)(a^k h)^4} \quad (19)$$

Since this software is aimed at computing the electric field derivatives of non-linear optical properties, *i.e.* the electronic energy, the dipole moment, the polarizability matrix, the first hyperpolarizability tensor, and so on, the definitions of the just stated properties read as

$$\mu_i := -\frac{\partial E(\mathbf{F})}{\partial F_i} \quad (20)$$

$$\alpha_{ij} := -\frac{\partial^2 E(\mathbf{F})}{\partial F_i \partial F_j} = \frac{\partial \mu_i}{\partial F_j} \quad (21)$$

$$\beta_{ijk} := -\frac{\partial^3 E(\mathbf{F})}{\partial F_i \partial F_j \partial F_k} = \frac{\partial^2 \mu_i}{\partial F_j \partial F_k} = \frac{\partial \alpha_{ij}}{\partial F_k} \quad (22)$$

$$\gamma_{ijkl} := -\frac{\partial^4 E(\mathbf{F})}{\partial F_i \partial F_j \partial F_k \partial F_l} = \frac{\partial^3 \mu_i}{\partial F_j \partial F_k \partial F_l} = \frac{\partial^2 \alpha_{ij}}{\partial F_k \partial F_l} = \frac{\partial \beta_{ijk}}{\partial F_l} \quad (23)$$

Hence, the derivatives coming minus-derivatives of the electronic energy define the non-linear optical property. As a consequence, the derivatives Eqs.(19) when using the energy must be negative. The Romberg procedure from Eq.(11).

2 Prerequisites of RomberG.exe

RomberG.exe is a FORTRAN95 flag-based script. FORTRAN95 is not capable of reading flags alone, that is why this code has been built using this module. Therefore, the first requirement is having installed, or downloaded, in your working directory before compiling RomberG.f95.

Next is compiling RomberG.f95 along with the getopt module with the following command line:

```
$ gfortran f90getopt.F90 -ffree-line-length-none -o ROMBERG.exe ROMBERG.f95
```

The `-ffree-line-length-none` compiling flag is mandatory - looking at the source code the reader is going to see that it clearly surpasses the limit of 80 characters per line. Once executed the previous command line, you are going to have a ROMBERG.exe in your working directory.

Moreover, it is necessary to have all fchk files of the molecule in a directory named after the molecule. It is recommended to name the fchk files as: *basename_field-direction_magnitude-of-the-field*. For instance, a valid fchk filename is "mol3_X-0.0004" or "ethaney0.0064".

If the user desires to generate RomberG-ready .fchk files without any trouble, there is also the 'makeromberg.py' Python script. This script works under the assumption that there is a com and log file in the directory under this name-template 'sp(name-of-the-molecule)0.com' with the keyword 'Field=X+000':

```
%nproc=4
%mem=8GB
%chk=spH2O0.chk
%oldchk=H2O.chk
# B3LYP/6-311+G(d,p) Polar NoSymm guess=read geom=check Field=X+000
...
```

This simple example uses as the name of the molecule 'H2O'. For further details, the user may read the 'makeromberg' script.

Furthermore, the latest iteration of RomberG is prepared for reading single-column text files. This way longitudinal properties can be computed by *grepping* the desired input property, *i.e.* the α_{zz} property for a series of explicit-field dependent calculations.

2.1 System-based prerequisites

- RomberG.exe is strictly prepared to read the Gaussian16 fchk files.
- Wherever this code is executed, please check the location of the grep command. Depending on whether RomberG.exe is executed in a cluster (HPC) it may be located in '/bin/grep' or in a personal laptop/desktop in '/usr/bin/grep'. For the latter case, also consider which FORTRAN compiler is being used: it has been tested for 'GNU Fortran (Debian 4.4.7-2) 4.4.7' and 'GNU Fortran (GCC) 4.8.5 20150623 (Red Hat 4.8.5-36)'.

Either way, RomberG.exe is prepared to work on both possibilities of grep location.

- Although the code is rather robust at reading the properties and "ignore" the file names, it is strongly encouraged that the folder that contains the .fchk files is the name of the molecule and the name of the .fchk files are named after something such as 'file' to avoid any kind of conflicts.

3 Execution of RomberG.exe and other optionalities

The optionalities of RomberG.exe can be displayed executing the following command line:

```
$ ./ROMBERG.exe -h {name_of_the_molecule}
```

and it is going to print in screen the following:

Several optionalities are available for RomberG.exe:

- i, -input : Input property - Energy/energy/E ; Dipole/dipole/M ; Alpha/alpha/A ; Beta/beta/B
- o, -output : Output property - Dipole/dipole/M ; Alpha/alpha/A ; Beta/beta/B ; Gamma/gamma/G
- F, -totalfields : Number of total fields probed. Integer required.
- I, -isotropic : Whether to compute isotropic properties or not.
- S, -simple : Simplifies the output to print properties under 10^{-6} with Romberg errors lower than 10% than its value, *i.e.* errors under 10^{-7} .
- p, -printP : Print the derivatives when computing the Romberg triangle for each component.
- Q, -init : Different first field. Default 1.0d-4. Real number expected.
- Y, -sqrt : Activate the step of $\sqrt{2}$ to compute the derivatives.
- J, -force : Force the complete calculation of isotropic properties despite non-computed values. The latter are approximated to zero.
- h, -help : Displays this message.

REMEMBER: The execution of RomberG.exe requires the name of the molecule, even for displaying this message.

For instance, a regular command line execution could be

```
$ ./ROMBERG.exe -i alpha -o gamma -F 13 -S -T {name_of_the_molecule}
```

this exemple of command line is going get the elements of the polarizability matrix of *.fchk, compute the elements of γ by computing the second derivative of each element of α with respect to X, Y, and Z. With those values it is going to print them in screen, compute several quantities related to the second hyperpolarizability and simplify the output displaying zeroes when necessary.

4 Output of RomberG.exe

RomberG.exe is designed not only to compute the formulae on Eqs.(19) but also computing isotropic and longitudinal-based properties.

Without specifying either the flags of isotropic and/or longitudinal, only the Romberg triangles are going to be printed along with the tensors composed with the propoerties with the minimum Romberg error using the Gaussian's fchk format.

Specifying the longitudinal flag enables the comutation of the following properties:^[2]

$$\mu = \sqrt{\mu_x^2 + \mu_y^2 + \mu_z^2} \quad (24)$$

$$\alpha = \frac{\alpha_{xx} + \alpha_{yy} + \alpha_{zz}}{3} \quad (25)$$

And prints the respective output tensor following Gaussian's format-checkpoint format. Toggling the isotropic flag activates the computation of the two previous equations and the ones below. All of them are rederived according following permutation symmetry:^{[3],[4],[5]}

$$\Delta\alpha = \frac{\sqrt{2}}{2}((\alpha_{xx} - \alpha_{yy})^2 + (\alpha_{yy} - \alpha_{zz})^2 + (\alpha_{zz} - \alpha_{xx})^2 + 12(\alpha_{xy}^2 + \alpha_{yz}^2 + \alpha_{xz}^2)) \quad (26)$$

$$\beta_{vec} = \sqrt{\sum_i^{x,y,z} (\sum_j^{x,y,z} \beta_{ijj})^2} \quad (27)$$

$$\beta_{||} = \frac{3}{5\mu} \sum_{i,j}^{x,y,z} \mu_i \cdot \beta_{ijj} \quad (28)$$

$$\beta_{\perp} = \frac{1}{5} \sum_{x,y,z}^{i,j} \mu_i \cdot (2\beta_{ijj} - 3\beta_{jij} + 2\beta_{jji}) \quad (29)$$

Notice that according to the symmetry constraints the code is subjected to Eqs.(28) and (29) are equivalent. And for γ the following quantity is computed:

$$\gamma_{||} = \frac{1}{5} \sum_{i,j}^{x,y,z} \gamma_{iijj} \quad (30)$$

When outputting either β or γ from properties of lower order, *i.e.* the electronic energy, the dipole moment or the polarizability, the analytic properties are fetched from the .fchk and the analytical dipole moment (Eq.(24)) and isotropic polarizability (Eq.(25)) are computed.

Since the code is open-source if the user wants to implement other formulae other than the ones presented here or correct the multiplying factors, they are free to modify the original code. For further corrections, implementations, corrections, requests, etc., feel free to contact through the email at the beginning of the document.

5 Summary of top-to-bottom execution of RomberG.exe

This section is a summary that covers from the optimization of the interest molecule to the execution of RomberG.exe

1. (*Recommended*) Optimization of the molecule. Use the basename of the molecule, *i.e.* H2O or CH3OH, for the following steps. It is also recommended to move the all the optimization files, except for the .chk, elsewhere.
2. Prepare the 'Polar NoSymm Field=X+000' calculation according to the last textbox of Section 2 and following the name convention - 'sp(NAME)0.com'. You may not read from the previous checkpoint file.
3. Execute makeromberg.py. For instance, \$ python makeromberg.py H2O .
4. Submit the calculations to your cluster.
5. In the parent directory of the molecule calculations, compile RomberG.f95 as stated before and execute following the guidelines presented in Section 3.
6. You may delete the checkpoint files to save on disk space, even the submission files can also be deleted. As long as all the necessary .fchk are present, all other files can be deleted.
7. Execute your desired command line of RomberG.

6 Possible conflicts with RomberG.exe and fchk files

Since this script is based to read from Gaussian fchk files, the computation of NLOPs at the DFT level is analytical. However, computing them at post-HF level is analytical only until the polarizability matrix. Consequently, toggling as input NLOP the first hyperpolarizability in this second scenario will return gibberish.

7 Bibliography

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